Shashwat Jaguri

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Education

University of Michigan — Ann Arbor, MI
B.S.E. in Computer Engineering, GPA: 3.62 — *Expected Grad: May 2027*
M.S.E in Electrical & Computer Engineering (SUGS Program) — *Expected Grad: May 2028*

Relevant Courses: Computer Architecture (EECS 470), VLSI Design (EECS 427), Digital Integrated Circuits, Digital Logic Design, Programming & Data Structures, Signals & Systems

Experience

Instructional Grader for EECS 470 — August 2026 – Present
CSE Department, University of Michigan — *Ann Arbor, MI*

- Graded homework, lab assignments, and exams on superscalar out-of-order processor design, memory hierarchies, and hardware verification for advanced computer architecture.

Club President — April 2026 – Present
RISC-M, University Student Organization — *Ann Arbor, MI*

- Co-founded UofM's first VLSI/Computer Architecture club; oversaw operations, secured corporate sponsorships, organized technical events, and directed student hardware design teams.

Undergraduate Research Assistant — May 2026 – September 2026
M-Bits Lab, University of Michigan | Advisor: Prof. Reetuparna Das — *Ann Arbor, MI*

- Built a custom C++ gem5 SimObject modeling Intel AMX, matching real Sapphire Rapids hardware within 6% while remaining fully behaviorally correct, establishing a cycle-accurate baseline for evaluating memory-side inference accelerators.
- Wrote optimized GEMM kernels in C using AMX intrinsics that outperformed Intel's oneDNN by a geomean of 7.8%, used to validate simulation accuracy against real workloads.
- Profiled Sapphire Rapids CPUs with targeted microbenchmarks to measure AMX latency and throughput, calibrating the simulator against empirical hardware data.

Projects

Superscalar Out-of-Order RISC Processor and Toolchain

- Designed and synthesized a 2-way superscalar, 5 stage out-of-order processor in a 230nm library, reducing average time per instruction by 38% ($19.7\ \text{ns} \rightarrow 12.14\ \text{ns}$) over a 3-stage in-order baseline.
- Engineered a micro-op queue splitting memory instructions into address-generation and memory-access operations, decoupling execution from the front-end.
- Integrated a gshare branch predictor, 2-way banked BTB, and set-associative I-cache to reduce front-end stalls.
- Verified the micro-op queue with standalone SystemVerilog testbenches targeting structural and handshake hazards; validated full-core correctness via architectural state trace diffing and waveform debugging.

Custom 16-bit RISC Processor Layout

- Designing a full-custom 16-bit RISC processor in the TSMC 65nm PDK as part of a 5-person team.
- Completed full-custom layout of the register file and ALU, verified with DRC and LVS in Mentor Graphics Calibre.
- Characterized block performance through transient simulations in Cadence Spectre and HSPICE.

LiDAR Data Processing Accelerator

- Built a real-time pipeline streaming VLP-16 LiDAR packets from custom HPS C drivers across the HPS-to-FPGA bridge, performing coordinate transformation, filtering, and occupancy grid mapping in hardware at 300k points/sec.
- Implemented a banked, double-buffered BRAM grid featuring read-modify-write updates, data forwarding, reduction logic, and structural hazard resolution for concurrent memory accesses.

Technical Skills

Languages: SystemVerilog, Verilog, C, C++, Python, MATLAB, Assembly (ARMv8, x86)
Tools & Technologies: Synopsys, Verdi, Cadence Tools, Intel Quartus, Git, Linux, PyTorch, ModelSim, PSpice, Altium Designer, STM32, gem5, HDL